

Amrit Sinha

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EDUCATION

BITS Pilani

B.E. (Hons.) Electronics and Instrumentation

2020 – 2024

EXPERIENCE

Instahyre

March 2025 – June 2026

SDE 1

- **Tech Stack:** Python, Django, MySQL, Redis, RabbitMQ, Elasticsearch, React, Jinja, HTML, CSS
- Built a **TypeScript MCP studio server** integrating AI reviewers with Phabricator, automating review-context generation from Differential diffs, changed files, and linked tasks.
- Integrated **AI coding assistants (Claude Code, GPT-Codex)** into the development workflow to automate boilerplate generation and feature scaffolding, enabling faster iteration and improved developer productivity.
- Implemented a **Redis-based sliding window rate limiter** using Lua scripts for atomic execution, throttling outbound emails to comply with AWS SES limits.
- Optimized **high-traffic API endpoints** with **p99 latencies exceeding 1000ms**, reducing them to under **300ms**.
- Designed and implemented a **Redis-backed OTP verification flow**, preventing premature database updates and ensuring secure phone number and email changes.
- Led a **large-scale refactor** to modularize the candidate-side application, reducing code coupling and accelerating feature development.
- Built and upgraded **UI components** using Jinja Templates, HTML, and custom CSS, improving responsiveness and maintainability.

ofScale

Nov 2024 – Jan 2025

Full Stack Engineer Intern

- **Tech Stack:** JavaScript, React.js, Next.js, Tailwind CSS, Golang, MySQL
- Developed and implemented the **mobile view** for the entire admin management website.
- Developed a **cron scheduler in Go** utilizing the gocron library, from initial design to final implementation.

Zluri Technologies

Jul 2023 – Dec 2023

Full Stack Engineer Intern

- **Tech Stack:** JavaScript, Node.js, React.js, Next.js, Express.js, MongoDB, PostgreSQL
- Built a **currency conversion feature** leveraging the Exchange Rate API to provide real-time and historical exchange rates.
- Migrated APIs to a new database architecture, reducing **response times by nearly 40%**.

PROJECTS

Multimodal RAG Backend | Python, FastAPI, Celery, Redis, PostgreSQL, Docker

2026

- Built a **document Q&A backend** (FastAPI) for PDF/image uploads with RAG and page-level citations.
- Designed an **async ingestion flow** with presigned MinIO uploads and a **background Celery worker** that indexes files with pollable ingestion job status and retries.
- Implemented **text and image retrieval** over Postgres + pgvector using **BGE** for passages and **CLIP** for images, feeding matched sources to **Gemma4 (12b)** via **Ollama** for grounded, cited answers.
- Added **SSE streaming**, optional API-key auth, Redis rate limiting, and Docker Compose deployment.

TECHNICAL SKILLS

Programming Concepts: Data Structures, Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, System Design (HLD, LLD)

Languages: JavaScript, TypeScript, Golang, Java, Python, C, C++, HTML5, CSS3, SQL

Frameworks & Libraries: React.js, Next.js, Redux, Angular, Node.js, Express.js, Bun.js, Django, Django Rest Framework, FastAPI, Celery, Tailwind CSS, Bootstrap, Jinja

Databases, & Messaging: PostgreSQL, MySQL, MongoDB, Elasticsearch, pgvector, Redis, RabbitMQ, Kafka

Tools & AI: Git, Linux, Docker, Kubernetes, AWS, Figma, Mercurial, Claude Code, GPT-Codex, RAG, MCP